C 40602 **(Pages : 3)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS–UG)

Mathematics

MTS 6B 10—REAL ANALYSIS

(2019 Admission onwards)

Time : Two Hours and a Half Maximum : 80 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks.*
Maximum marks 25.

1. State sequential criterian for continuity.
2. Show that the sine function is continuous on $\mathbb{R}$.
3. Define Lipchitz function. If $f : A \to \mathbb{R}$ is a Lipschitz function then show that f is uniformly continuous on A.
4. Define tagged partition.
5. Show that every constant function on $[a,b]$ is in $\mathbb{R}[a,b]$.
6. State Cauchy's criterion for Riemann integrability.
7. Let F, G be differentiable on $[a,b]$ and let $f = F'$ and $g = G'$ belong to $\mathbb{R}[a,b]$, then show that $\int_a^b f\,G = [FG]_a^b - \int_a^b F\,g$.
8. Show that $\lim\left(\frac{x}{n}\right) = 0$ for $x \in \mathbb{R}$.
9. Define uniform convergence of a sequence of functions.
10. State bounded convergence theorem.
11. State Weirstrass M-test for the uniform convergence of series of functions.
12. Evaluate $\int_1^\infty \frac{dx}{x^2+1}$.

Turn over

13. Find the principal value of $\int_{-2}^{3} \frac{dx}{(x-1)^3}$.

14. Discuss the absolute convergence of $\int_0^{\infty} \frac{\sin x}{n+1}\, dx$ for $n\pi \le x \le (n+1)\,\pi,\ n = 0,1,2,\ldots$

15. If $\int_0^b \frac{dx}{1+ax} = \frac{1}{a}\ln(1+ab)$. Evaluate $\int_0^b \frac{x\,dx}{(1+ax)^2}$.

Section B

Questions 16–23, *answer any number of questions.*
Each question carries 5 *marks.*
Maximum marks 35.

16. State and prove Boundedness theorem for continuous function.

17. Show that $f(x) = \frac{1}{1+x^2}, x \in \mathbb{R}$ is uniformly continuous in $\mathbb{R}$.

18. State and prove Squeeze theorem for Riemann integrable functions.

19. If $f \in \mathbb{R}[a,b]$ and f is continuous at a point $c \in [a,b]$. Then show that the indefinite integral $\mathrm{F}(z) = \int_a^z f$ for $z \in [a,b]$ is differentiable at c and $\mathrm{F}'(c) = f(c)$.

20. Show that a sequence (f_n) of bounded functions on $\mathrm{A} \subset \mathbb{R}$ converges uniformly on A to f iff $\| f_n - f \| n \to 0$.

21. Discuss the convergence of $f_n(x) = \frac{x^n}{n+x^n}, x \ge 0$. Is the convergence uniform on $[0,\infty]$.

22. Evaluate $\int_{-1}^{1} \frac{dx}{x^2-1}$.

23. Show that $\forall\ q \in \mathbb{R}, \int_1^{\infty} x^q\, e^{-x}\, dx$ converges.

Section C

Questions 24–27, *answer any* **two** *questions.*
Each question carries 10 *marks.*

24. State and prove Maximum Minimum Theorem.

25. State and prove Cauchy's criterion of Riemann integrability.

26. Let (f_n) be a sequence of functions in $\mathbb{R}[a,b]$ and suppose that (f_n) converges uniformly on $[a,b]$ to f. Then show that $f \in \mathbb{R}[a,b]$.

27. Show that $\int_0^\infty \frac{\sin x}{x} dx$ exists and converges to a finite real value and that this integral does not converge absolutely.

(2 × 10 = 20 marks)

C 40602-A **(Pages : 4)** **Name.....**

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SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS–UG)

Mathematics

MTS 6B 10—REAL ANALYSIS

(2019 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 20**	**Maximum : 20 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 10—REAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Choose the correct option :

   (A) Let $a, b \in \mathbb{R}$ and $f : (a, b] \to \mathbb{R}$ continuous on $(a, b]$ then f has an absolute maximum and absolute minimum on $(a, b]$.

   (B) Let $a, b \in \mathbb{R}$ and $f : [a, b] \to \mathbb{R}$ continuous on $[a, b]$ then f has an absolute maximum and absolute minimum on $[a, b]$.

   (C) Let $a, b \in \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ continuous on (a, b) then f has an absolute maximum and absolute minimum on (a, b).

   (D) Let $a, b \in \mathbb{R}$ and $f\,[a, b) \to \mathbb{R}$ continuous on $[a, b)$ then f has an absolute maximum and absolute minimum on $[a, b)$.

2. Consider the statements :

   I : If f is a real valued uniformly continuous function on (a, b) then there exists a continuous extension of f to $[a, b]$.

   II : If f is a real valued continuous function on (a, b) then there exists a continuous extension of f to $[a, b]$.

   (A) Both I and II are true. (B) I is false and II is true.

   (C) I is true and II is false. (D) Both I and II are false.

3. The values of x for which the function $\dfrac{x^2 - 3x - 4}{x^2 + 3x - 4}$ Which of the following is correct ?

   (A) $f(x)$ is continuous at $x = 1$. (B) $f(x)$ is continuous at $x = 4$.

   (C) $f(x)$ is continuous at $x = -4$. (D) $f(x)$ is continuous at $x = 0$.

4. If $a^2 + b^2 + c^2 = 1$ then $ab + bc + ac$ lies in the interval :

   (A) $\left[\dfrac{1}{3}, 2\right]$. (B) $\left[\dfrac{-1}{2}, 1\right]$.

   (C) $\left[-1, \dfrac{1}{2}\right]$. (D) $[0, 2]$.

5. The value of $\lim\limits_{x \to 0} (\tan x \log x)$:

   (A) 2. (B) 1.

   (C) 3. (D) 0.

6. The value of $\lim\limits_{x \to \infty} \dfrac{x \ln x}{1 + x^2}$ is :

   (A) 0. (B) 1.

   (C) e. (D) ∞.

7. For the function e^{-x}, the linear approximation around $x = 2$ is :

(A) $(3 - x)e^{-2}$. (B) $1 - x$.

(C) $\left[3 + \sqrt{2} - 1\left(1 + \sqrt{2}\right)x\right]e^{-2}$. (D) e^{-2}.

8. Consider the dollowing statements :

I : Let $f(x) := 2$ if $0 \le x < 1$ and $f(x) := 1$ if $1 \le x \le 2$. Then $f \in \mathcal{R}[0, 2]$.

II : Let $h(x) := 2$ if $0 \le x < 1$, $h(1) := 3$ and $h(x) := 3$ and $h(x) := 1$ if $1 \le x \le 2$. Then $h \in \mathcal{R}[0, 2]$.

Choose the true statement :

(A) Both I and II are true. (B) I is true and II is false.

(C) I is false and II is true. (D) Both I and II are false.

9. If $f_1, f_2, f_3, ..., f_n$ are in $\mathcal{R}[a, b]$ then :

(A) $\frac{f_i^2}{f_j} \in \mathcal{R}[a, b]$ for all $i, j = 1, 2, ..., n.$ (B) $f_i - f_j \in \mathcal{R}[a, b]$ for all $i, j = 1, 2, ..., n.$

(C) $\frac{f_i}{f_j} \in \mathcal{R}[a, b]$ for all $i, j = 1, 2, ..., n.$ (D) $\frac{f_i - f_2}{f_j} \in \mathcal{R}[a, b]$ for all $i, j = 1, 2, ..., n.$

10. Find $\lim(\text{Arctan } nx)$ for $x \in \mathbb{R}$:

(A) $(\pi/2)\,\text{sg}\, nx.$ (B) $\pi\,\text{sg}\, nx.$

(C) $2\pi\,\text{sg}\, nx.$ (D) $0.$

11. Find $\lim\left(xe^{-nx}\right)$ for $x \in \mathbb{R}, x \ge 0$:

(A) $-1.$ (B) $0.$

(C) $1.$ (D) $2.$

12. Choose improper integral from below :

(A) $\int_0^{\infty} \sin(x)\, dx.$ (B) $\int_0^{1} \sin(x)\, dx.$

(C) $\int_0^{2} \sin(x)\, dx.$ (D) $\int_0^{3} \sin(x)\, dx.$

13. Find the value of improperintegral $\int_{-1}^{0} \frac{dx}{\sqrt[3]{x}}$:

(A) $\frac{-1}{2}.$ (B) $\frac{1}{2}.$

(C) $\frac{-3}{2}.$ (D) $\frac{3}{2}.$

14. The integral $\int_a^b \frac{dx}{x^2-1}$ is :

(A) Improper when $a=-1$.
(B) Improper when $b=-1$.
(C) Improper when $b=1$.
(D) All of these.

15. Choose the true $\int_{-1}^{1} \frac{\sin(x)}{x}\,dx =$

(A) $\sum_{n=0}^{\infty} (-1)^n \frac{2}{(2n+1)(2n+1)!}$.
(B) $\sum_{n=0}^{\infty} (-1)^n \frac{2}{(2n)(2n)!}$.
(C) $\sum_{n=0}^{\infty} (=1)^n \frac{2}{(2n+1)(2n)!}$.
(D) $\sum_{n=0}^{\infty} (-1)^n \frac{2}{(2n)(2n+1)!}$.

16. Find P.V. $\int_{-\infty}^{\infty} x dx$:

(A) 1.
(B) ∞.
(C) 0.
(D) $-\infty$.

17. Choose the true statement :

(A) $\int_{-2}^{3} \frac{dx}{x} = \ln\left(\frac{3}{2}\right)$ and P.V. $\int_{-2}^{3} \frac{dx}{x} = \ln\left(\frac{3}{2}\right)$.
(B) $\int_{-2}^{3} \frac{dx}{x}$ does not exist and P.V. $\int_{-2}^{3} \frac{dx}{x} = \ln\left(\frac{3}{2}\right)$.
(C) $\int_{-2}^{3} \frac{dx}{x}$ does not exist and P.V. $\int_{-2}^{3} \frac{dx}{x}$ does not exist.
(D) $\int_{-2}^{3} \frac{dx}{x}\frac{3}{2}$ and P.V. $\int_{-2}^{3} \frac{dx}{x} = \ln\left(\frac{3}{2}\right)$.

18. P.V. $\int_{-\infty}^{\infty} \frac{dx}{x}$

(A) 1.
(B) ∞.
(C) 0.
(D) Does not exist.

19. Find P.V. $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$:

(A) π.
(B) $\frac{\pi}{2}$.
(C) 0.
(D) $-\pi$.

20. Let 0 denotes composition of two functions. Also let even stands for even function and odd stands for odd function, then :

(A) even 0 even = odd.
(B) even 0 odd = odd.
(C) odd 0 even = odd.
(D) odd 0 odd = odd.